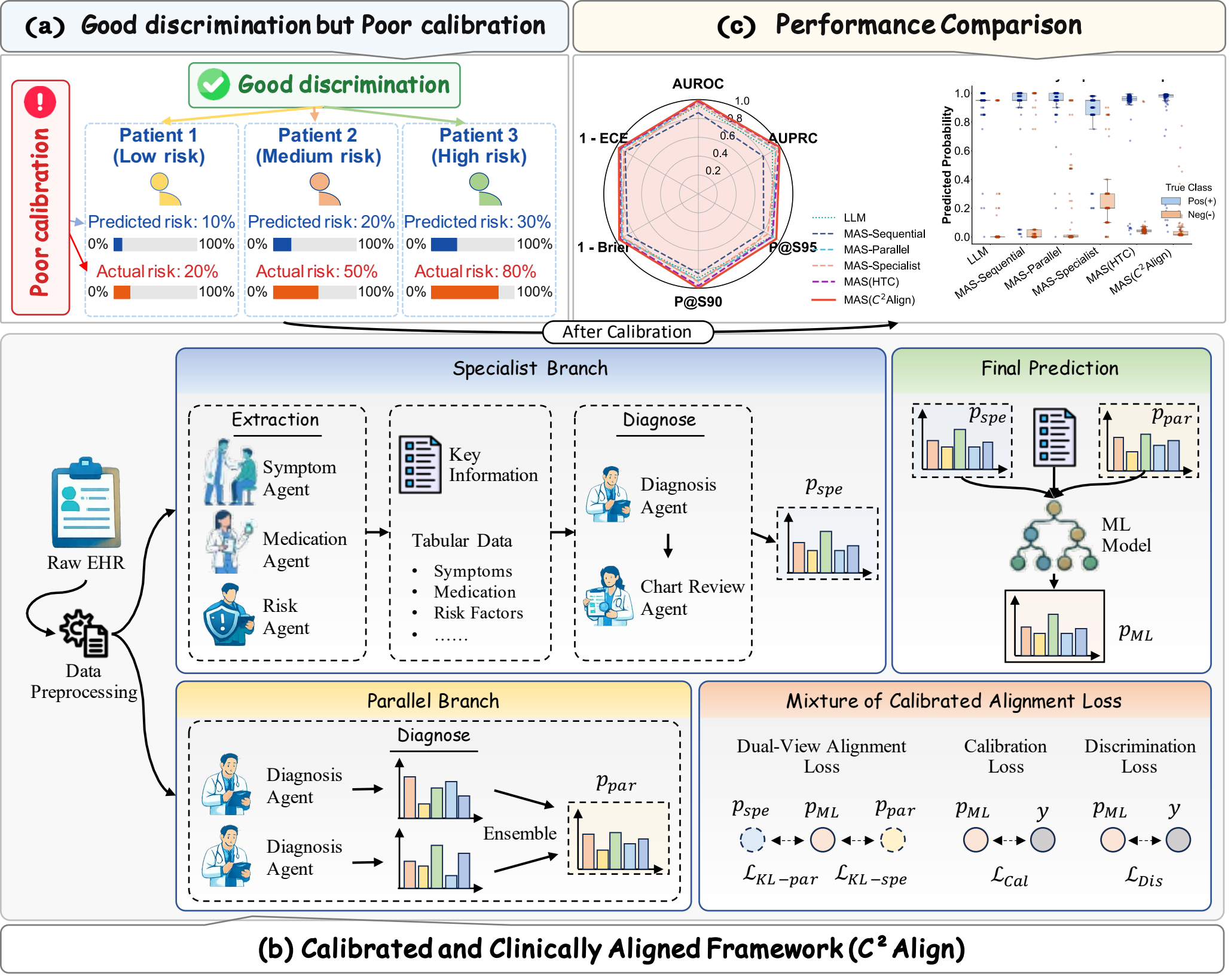

(a) Good discrimination but Poor calibration
Poor calibration
Good discrimination
Patient 1 (Low risk)
Patient 2 (Medium risk)
Patient 3 (High risk)
Predicted risk: 10%
Predicted risk: 20%
Predicted risk: 30%
0%
100%
Actual risk: 20%
Actual risk: 50%
Actual risk: 80%
(c) Performance Comparison
AUROC
AUPRC
P@S95
P@S90
1 - Brier
1 - ECE
LLM
MAS-Sequential
MAS-Parallel
MAS-Specialist
MAS(HTC)
MAS(C²Align)
Predicted Probability
True Class
Pos(+)
Neg(-)
After Calibration
Specialist Branch
Extraction
Symptom Agent
Medication Agent
Risk Agent
Key Information
Tabular Data
Symptoms
Medication
Risk Factors
……
Diagnose
Diagnosis Agent
Chart Review Agent
p_{spe}
Final Prediction
p_{par}
ML Model
p_{ML}
Raw EHR
Data Preprocessing
Parallel Branch
Diagnose
Diagnosis Agent
Diagnosis Agent
Ensemble
p_{par}
Mixture of Calibrated Alignment Loss
Dual-View Alignment Loss
Calibration Loss
Discrimination Loss
p_{spe} p_{ML} p_{par}
p_{ML} y
p_{ML} y
$\mathcal{L}_{KL-par}$ $\mathcal{L}_{KL-spe}$
$\mathcal{L}_{Cal}$
$\mathcal{L}_{Dis}$
(b) Calibrated and Clinically Aligned Framework (C² Align)